Monday (EDL-13)

Group code: -12 **Student names:** P. Ramya, Oviya S, Shiwani, Khushbu, Kanak **EDL 2025**

Project title: P-21 Ground Penetrating Radar **Date:**

Use your notebooks for discussions and rough work. Fill out this sheet after working individually and discussing within your team.

1. **In simple words, describe what you are going to build in your project, what its purpose is, and how it will function.** Be as detailed as possible, covering all the major aspects of your project.
   a. What is the main goal of your project?
   b. What problem does it solve, and how?
   c. Who will use your project, and in what context?

   Draw a pencil sketch of what your project will look like at the end of the course, for final demo.

a. to design and develop a system capable of detecting and visualising subsurface structures using radio waves - transception.

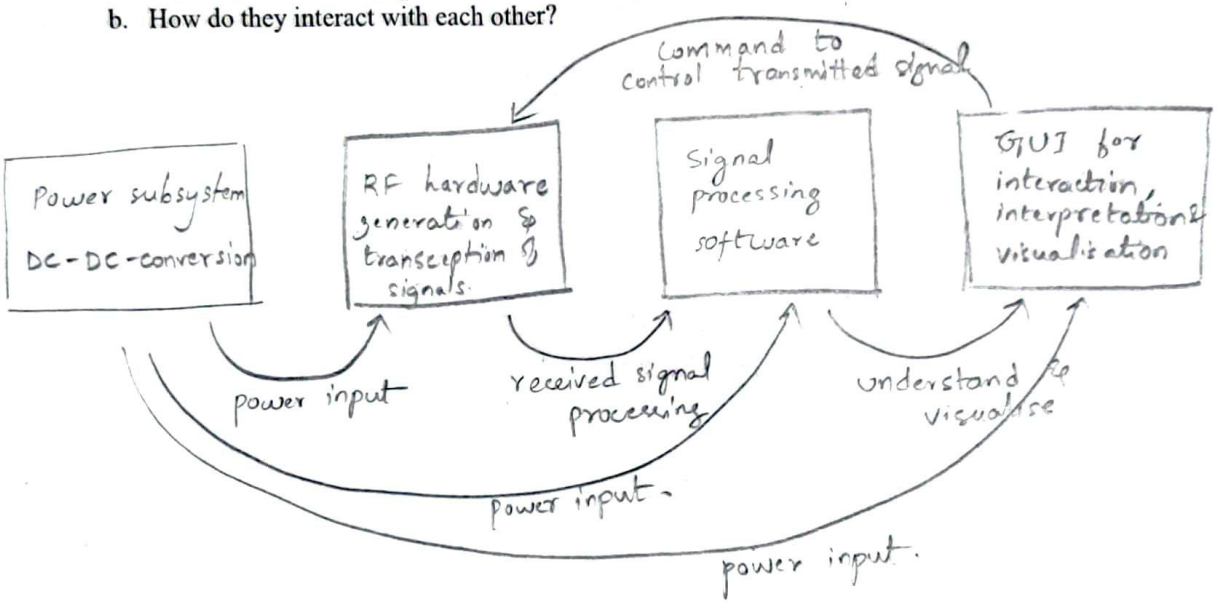
b. Our project solves the problem of non-invasive subsurface exploration without the need for excavation or drilling.

c. How- leveraging the penetrating and reflecting properties of radio waves to accurately & detect and visualise subsurface objects and structures.

c. Our project can be used by a variety of professionals and industries.

Eg. 1. Construction and Civil engineering
2. Archaeologists.
3. Geologists and Environmentalists
4. Military and Security application
5. Infrastructure maintenance.
6. Rescue operations.

2. **Draw a block diagram of your project.** Create a visual representation showing the key components or subsystems of your project. For each block in the diagram, briefly explain its main function and how it fits into the overall system.
   a. What are the main subsystems or modules of your project?
   b. How do they interact with each other?



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3. **Write down details for these blocks:** What are the key performance metrics for each block (e.g., power, size, speed)? What trade-offs are you considering in your design choices? Are there any constraints or limitations for each block?

Block	Key specifications of this block	Design choices for this block
RF Hardware	* SNR – especially for the reflected signal, amplifying and removing noise * Mixer fn – to accurately split the signal for reference calibration * Transmitted signal strength – how to make a balance b/w power & strength of the signal to penetrate	→ LNA, Splitter – already built-in modules are available
DC-DC conversion	* Switching frequency: - to control the size of the circuitry and achieving the required voltage.	→ Boost converter
Software	- Dynamic / Smooth user control - Signal processing & visualization: To clearly read the hyperbolas & drawing the most approximate visuals.	→ Python PyQT5 for GUI development

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4. **What are the unknowns or uncertainties in this project?** Identify aspects of your project that you are uncertain about or that require further research. This may include areas where you know what you need to do but are unsure how to approach it.
    a. What technical challenges or questions are you facing?
    b. Are there any assumptions you must make in order to move forward?

a) • Working with RFs might prove as a challenge. Integration with SDR for signal processing is going to be tough.
• Creating a full-fledge GUI along with the feature to generate report & further. real-time, how to make the display stand-alone without compromising on the functionalities.

b) • Currently, we are relying on GNU-Radio Companion as it is a readily used FOSS.
• For displays, we are assuming that we can do the visualization & processing using GNU on a tablet.

Other things to consider from now until Milestone 1 deadline:

5. **Roles and Responsibilities: How will the work be divided among team members?** Assign specific tasks and responsibilities to each team member. Be clear about who is responsible for each part of the project.
    a. Who will work on which blocks or subsystems?
    b. What are the deadlines for each task?
    c. How will the team communicate and coordinate to ensure everyone is on track?

6. **Next Steps: What is your plan for the next phase of the project?** Outline what needs to be done in the short-term to move forward.
    a. What are the immediate next tasks or priorities?
    b. Are there any dependencies between tasks? How will you handle these interdependencies?
    c. What resources or materials do you need to proceed?

7. **Feedback and Collaboration: How will you gather feedback and collaborate during the project?** Describe how your team plans to share progress, give and receive feedback, and collaborate throughout the course of the project.
    a. How often will you check in with your team members?
    b. Will you conduct regular brainstorming or review sessions?

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